CODEFUSION: PYTHON & JAVA SHOWDOWN
PYTHON & JAVA-BASED CODING COMPETITION
Round 2 - Scenario-Based Questions (CloudCoder Platform)

Problem 1: Secure Password Validator

Scenario:
A system requires passwords that meet the following security criteria:

- At least **8 characters long**
- Contains at least **one uppercase letter**
- Contains at least **one number**
- Contains at least **one special character** (@, #, $, %, &, *)

Expected Input:

- A string (password)

Expected Output:

- "Valid Password"
- "Invalid Password"

Example:

Input: P@ssw0rd
Output: Valid Password
Input: password123
Output: Invalid Password

Problem 2: Flight Ticket Pricing System

Scenario:
An airline charges different ticket prices based on passenger age:

- **Children (0-12 years) → 50% discount**
- **Senior Citizens (60+ years) → 30% discount**
- **Standard fare otherwise**

Expected Input:

- An integer (ticket_price)
- An integer (age)

Expected Output:

- A single integer representing the final ticket price.

Example:

Input: 2000 10
Output: 1000
Input: 3000 65

Output: 2100

Problem 3: Banking Fraud Detection

Scenario:
A bank flags transactions as **fraudulent** if:

- The transaction **exceeds ₹50,000**
- The transaction occurs **outside normal banking hours (9 AM - 9 PM)**

Expected Input:

- An integer (transaction_amount)
- An integer (transaction_time)

Expected Output:

- "Transaction Approved"
- "Transaction Flagged as Fraud"

Example:

Input: 60000 23
Output: Transaction Flagged as Fraud
Input: 30000 15
Output: Transaction Approved

Problem 4: Warehouse Inventory Management

Scenario:
A warehouse automatically restocks an item if its quantity falls below 10.

Expected Input:

- An integer (current_stock)
- An integer (restock_quantity)

Expected Output:

- "Stock Sufficient"
- "Restocking: <restock_quantity>"

Example:

Input: 5 20
Output: Restocking: 20
Input: 15 30
Output: Stock Sufficient

Problem 5: Smart Traffic Light System

Scenario:
A **smart traffic system** dynamically adjusts green light duration based on traffic density:

- High Traffic (80%+ full) → Green light **60 sec**
- Medium Traffic (50%-79%) → Green light **40 sec**
- Low Traffic (<50%) → Green light **20 sec**

Expected Input:

- An integer (max_capacity)
- An integer (current_vehicle_count)

Expected Output:

- An integer representing green light duration in seconds.

Example:

Input: 100 85
Output: 60
Input: 100 45
Output: 20

Problem 6: Loan Eligibility Checker

Scenario:
A bank approves loans based on the following criteria:

- **Credit Score ≥ 700**
- **Monthly Income ≥ ₹25,000**
- **Existing Loan EMI ≤ 40% of Monthly Income**

Expected Input:

- An integer (credit_score)
- An integer (monthly_income)
- An integer (existing_loan_emi)

Expected Output:

- "Loan Approved"
- "Loan Rejected"

Example:

Input: 750 30000 8000
Output: Loan Approved
Input: 650 25000 5000
Output: Loan Rejected

Problem 7: Movie Recommendation System

Scenario:
A streaming platform recommends a movie based on **age and preference**:

- **Kids (≤ 12 years)** → "Animated Movies"
- **Teenagers (13-18 years)** → "Action Movies"
- **Adults (19+ years)** → "Drama Movies"

Expected Input:

- An integer (age)
- A string (preferred_genre)

Expected Output:

- Recommended movie category (string)

Example:

Input: 10 Animation
Output: Animated Movies
Input: 17 Action
Output: Action Movies

Problem 8: Fuel Station Discount System

Scenario:
A **fuel station** offers discounts based on fuel type:

- "Petrol" → ₹2 per liter discount if > 20 liters
- "Diesel" → ₹3 per liter discount if > 25 liters
- No discount otherwise

Expected Input:

- A string (fuel_type)
- An integer (quantity)

Expected Output:

- "Discount Applied: ₹<amount>"
- "No Discount"

Example:
Input: Petrol 25
Output: Discount Applied: ₹50
Input: Diesel 30
Output: Discount Applied: ₹90

Problem 9: Toll Booth Payment System

Scenario:
A toll booth charges vehicles based on type:

- "Car" - ₹50
- "Truck" - ₹100
- "Bike" - ₹20

Expected Input:

- A string (vehicle_type)

Expected Output:

- Toll Fee (int)

Example:

Input: Car
Output: 50
Input: Truck
Output: 100

Problem 10: E-commerce Shipping Cost Calculation

Scenario:
An e-commerce platform calculates shipping charges based on order weight:

- **Weight ≤ 5kg** → ₹50
- **5-20kg** → ₹100
- **Above 20kg** → ₹200

Expected Input:

- An integer (order_weight)

Expected Output:

- Shipping Cost (int)

Example:

Input: 3
Output: 50
Input: 12
Output: 100